

AWS DevOps – Comprehensive Enterprise Training Curriculum

Total Duration: 96 Hours (Instructor-Led Training) + 2-Week Capstone Extension. Hands-on Coverage: ~75%

Program Overview

Total Duration: 96 Hours

Delivery Mode: Instructor-led (Virtual / Onsite)

Focus Areas:

- DevOps principles and CI/CD pipelines on AWS
- Infrastructure as Code (CloudFormation & Terraform)
- Containerization and orchestration (Docker, ECS, EKS)
- GitOps and progressive delivery
- DevSecOps and compliance automation
- Monitoring, logging, and observability
- Multi-region deployments and cost optimization

Pre-requisite Knowledge

- Good understanding of Linux and networking fundamentals
- Familiarity with AWS console and core services
- Good scripting knowledge (Python or Bash preferred)
- Awareness of application deployment concepts

NOTE: Participants who do not meet the above prerequisites may require additional support to fully benefit from the hands-on and implementation-focused sessions.

Training Outcomes

By the end of this program, participants will be able to:

- Understand and apply core DevOps practices in AWS environments
- Build and manage CI/CD pipelines using AWS-native and third-party tools
- Implement Infrastructure as Code using CloudFormation and Terraform
- Containerize applications and deploy them using ECS and EKS
- Apply GitOps workflows and progressive deployment strategies
- Integrate security practices into DevOps pipelines (DevSecOps)
- Implement monitoring, logging, and observability solutions
- Work with multi-region deployment and cost optimization strategies

Training Methodology and Training Outcome

Training Methodology

- **Conceptual Learning:** ~20–25%
- **Guided Hands-on Labs:** ~55–60%
- **Capstone Implementation:** ~15–20%

Total Practical Exposure: ~75%

Assessment Strategy (Included in Duration)

- **Pre-Assessment (1 Hour)**
 - Conducted at the beginning
- **Module-wise:** Formative checks: end-of-module quizzes or short reflections.
- **Module-wise:** Practical assessments: completion of module labs.
- **Post-Assessment (1 Hour)**
 - Conducted at the end
 - Practical and scenario-based evaluation
- **Summative: final capstone project**

Module Duration & Hands-on Summary Table

Module	Topic	Total Hours	Hands-on Hours
Pre	Pre-Assessment	2	-
M1	DevOps & CI/CD Fundamentals	19	14
M2	Infrastructure as Code (IaC)	15	12
M3	Containerization & Registry	10	8
M4	Kubernetes with EKS	14	12
M5	GitOps & Deployment Patterns	8	6
M6	Security & Compliance (DevSecOps)	8	6
M7	Monitoring & Observability	10	8
M8	Advanced Topics & Production Readiness	8	6
Post	Post-Assessment	2	-

Total Duration: 96 Hours

Total Hands-on: ~72 Hours (~75%)

Detailed Curriculum

Module 1 – DevOps & CI/CD Fundamentals on AWS

Duration: 19 Hours (~14 Hours Hands-on)**

Topics Covered:

- DevOps principles, CALMS framework
- CI/CD concepts and Git strategies
- AWS CodeCommit, CodeBuild, CodeDeploy, CodePipeline
- Deployment strategies (Blue/Green, Canary, Rolling)
- GitHub Actions, GitLab CI, Jenkins integration
- Testing automation and quality gates

Hands-on Labs:

- Source control setup and Git workflows
- CI/CD pipeline implementation
- Multi-target deployment strategies
- Third-party CI/CD integration
- Automated testing and quality gates

Module 2 – Infrastructure as Code (IaC)

Duration: 15 Hours (~12 Hours Hands-on)**

Topics Covered:

- CloudFormation advanced concepts
- Terraform fundamentals and architecture
- Terraform modules and best practices
- IaC integration in CI/CD pipelines
- Policy as Code and compliance validation

Hands-on Labs:

- CloudFormation templates and StackSets
- Terraform provisioning and state management
- Module creation and reuse
- Automated IaC pipelines and security scanning

Module 3 – Containerization & Registry Management

Duration: 10 Hours (~8 Hours Hands-on)**

Topics Covered:

- Docker architecture and optimization

- Amazon ECR and container security
- ECS and Fargate deployment models
- Container orchestration decision patterns

Hands-on Labs:

- Docker image optimization and multi-stage builds
- Secure container pipeline with ECR
- ECS cluster and service deployment

Module 4 – Kubernetes with Amazon EKS

Duration: 14 Hours (~12 Hours Hands-on)**

Topics Covered:

- Kubernetes core concepts and architecture
- EKS setup and cluster management
- Networking, storage, and security in EKS
- Helm and scaling strategies

Hands-on Labs:

- Kubernetes deployments and services
- EKS cluster provisioning and configuration
- Advanced networking and storage setup
- Helm-based deployments and scaling

Module 5 – GitOps & Advanced Deployment Patterns

Duration: 8 Hours (~6 Hours Hands-on)**

Topics Covered:

- GitOps principles and workflows
- ArgoCD and Flux CD
- Progressive delivery strategies
- Canary and blue/green deployments

Hands-on Labs:

- ArgoCD setup and application deployment
- Flux CD configuration
- Progressive delivery implementation

Module 6 – Security & Compliance (DevSecOps)

Duration: 8 Hours (~6 Hours Hands-on)**

Topics Covered:

- DevSecOps principles
- AWS security services and IAM best practices
- Security automation in pipelines
- Zero Trust architecture and compliance frameworks

Hands-on Labs:

- Security pipeline integration
- IAM and secrets management
- Vulnerability scanning and compliance automation

Module 7 – Monitoring, Logging & Observability

Duration: 10 Hours (~8 Hours Hands-on)**

Topics Covered:

- CloudWatch monitoring and logging
- Distributed tracing with X-Ray
- OpenTelemetry, Prometheus, Grafana
- OpenSearch and centralized logging

Hands-on Labs:

- CloudWatch dashboards and alerts
- Distributed tracing implementation
- Centralized logging with OpenSearch

Module 8 – Advanced Topics & Production Readiness

Duration: 8 Hours (~6 Hours Hands-on)**

Topics Covered:

- Multi-region architectures and disaster recovery
- Route 53, Global Accelerator
- Backup strategies and failover patterns
- Cost optimization and FinOps practices

Hands-on Labs:

- Multi-region deployment and failover
- Backup and recovery implementation
- Cost optimization dashboards

Post-Assessment

Duration: 2 Hours

- Practical evaluation covering CI/CD, IaC, containers, and monitoring
- Scenario-based implementation questions

Capstone Project (Post-Training Phase)

Duration: 2 Weeks (Post-Training Extension)

Nature of Capstone

- Guided implementation of an end-to-end AWS DevOps solution
- Focus on integration of concepts rather than full production deployment

Scope (High-Level)

Participants will work on:

- CI/CD pipeline implementation
- Infrastructure provisioning using IaC
- Containerized application deployment (EKS/ECS)
- GitOps-based deployment workflow
- Monitoring, logging, and security integration
- Multi-environment setup and basic DR considerations

Note: Exact implementation depth may vary based on participant pace and team structure.

Post-Training Lab Access

Participants will be provided **extended lab access for 2 weeks post-training**, enabling continued practice and capstone implementation.

- Lab access will be available for up to standard working hours (approximately 7–8 hours per day)
- Access timings will be pre-scheduled and aligned with participant availability prior to the start of the capstone phase
- The scheduled window ensures availability of support team during active lab hours
- Multiple attempts (up to 4–5 iterations)

Instructor Support

- Up to 1 hour per day during lab window
- Provided by qualified instructor (same trainer may not always be available)
- Focused on guidance, not full re-implementation